Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/17/2004
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Solid Waste Resource Recovery		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	4
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Alleman , Jim (Purdue Univesity) Glass, Charles (Howard University) Volenec, Jeffrey (Purdue University) Joern, Brad (Purdue University) Brown, Paul (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The primary goal of this project is to develop a novel high-temperature solids digestion system for processing biodegradable wastes within a sustainable closed-loop ecosystem. The pathogen-free residuals streams generated by this process would be amenable to direct water and nutrient recovery. The design is loosely modeled after the successful, commercial Autothermal Thermophilic Aerobic Digestion (ATAD) wastewater sludge treatment process . Our NSCORT 'solid waste group' members have all maintained productive levels of individual research activity and collaborative interaction over the past year, and the critical process of securing timely distribution with representative STAR-product (both scrubber condensate and residual solids) has been now been reasonably well worked out in terms of delivery mode and sample volume. While prior collaborative visits to the Howard campus had been made during the course of our NSCORT operation, a particular highlight for FY2004 was a visit to Dr. Beyl's 'mushroom growth' lab at Alabama A&M. Admittedly, Dr. Beyl's particular project is considered a 'plant growth' center activity, but given the fact that STAR-residuals comprise a primary waste substrate with their mushroom growth an on-site visit was quite helpful in terms of comparing notes on overlapping research issues and concerns. STAR associated objectives are: 1- To create a thermophilic STAR reactor system mechanically able to suitable mix and aerate high input solids levels at an expected range of 6-10% solids, 2- To maximize STAR's positive ESM-enhancing attributes in terms of gas transfer rates to maximize performance, solids shear to reduce particle size, solids and organic destruction performance to reduce waste volume and enable further recovery of water, nutrients and carbon, and pathogen pasteurization to prepare the product for further use or storage (2,3), 3- To minimize STAR's ESM-degrading attributes, in regard to mass and volume, crewtime, noise generation, vibration, energy consumption, and heat loss, 4- To evaluate the STAR system's performance utilizing a long-term mission waste stream including human fecal, paper, food residuals, and plant materials, 5- To evaluate the complementary use and operational performance of a vacuum waste collection system, particularly in regard to high-solids waste transfer, and 6- To provide collaborative support for subsequent post-processing NSCORT research projects, including those of Charles Glass (off-gas air processing and condensate processing), Paul Brown (residuals uptake via fish), Jeff Volenec and Brad Joern (residuals water separation via plants), and Caula Behl (nutrient and water recovery via mushroom growth). Nitrogen Cycling in ALS: The primary objective of this research project is to effectively treat ammonia rich wastewaters from various process streams and convert it into a form that is amenable to uptake by plants in order to complete the nitrogen cycle throughout ALS. The sources of ammonia will include the condensate from off-gas from the STAR operation (on the order of 1,000 ppm of ammonium as nitrogen) and the urine freeze-thaw treatment system, which will also have an ammonium-rich brine discharge. The objective of effectively treating an ammonia rich wastewater will be met by constructing a zeolite system. At this point we believe that the use of a zeolite is the most efficient method to remove the concentrated ammonia from water, with the lowest equivalent system mass, and to remove the ammonia in a form that can be used as a possible input to plant growth systems inside NSCORT. Biosolids Dewatering: • Identify biomass specie(s) best suited for dewatering and capturing nutrients in STAR waste effluent. • Develop a cropping system that utilizes the dewatered biosolids form the STAR reactor to grow crops for food, mushroom growth and fish feed. • Investigate plant growth substrates for STAR effluent dewatering process. Waste Treatment Using Tilapia: • Evaluate ability of tilapia to reduce equivalent system mass • Evaluate quality of tilapia as food when fed waste products
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Further development of biological methods for converting organic wastes into fuels, chemicals, and other useful products. Add to knowledge about low-flush toilet operations
Task Progress:	1) STAR Solid Waste Processing (Jim Alleman @ Purdue) Extended testing with two different 'hydraulic retention times' (HRTs) (i.e., 11 and 16 days) and influent solids concentrations (i.e., 3 and 6%) was completed over the past year, and the latter, higher HRT run was conducted at a higher temperature (~60oC) such that sizably higher levels of solids degradation and fecal coliform loss were observed. In addition, an off-gas scrubber system was installed such that a realistic off-gas condensate stream could be obtained for subsequent evaluation with respect to zeolite sorption of ammonia. 2) STAR Off-Gas Processing (Charles Glass @ Howard): Dr. Glass's research team has completed an extensive series of batch zeolite sorption studies, including a series of runs with actual post-scrubber STAR off-gas condensate. In one such study, where there was no detectable NH3-N for approximately 40 bed volumes, a 'commendable' performance assessment appeared warranted given that BREATHe would not be expecting any significant incoming ammonia-nitrogen loading. 3) STAR residuals post-processing with Tilapia (Paul Brown @ Purdue): The accompanying photograph was taken during a site-visit to Purdue's aquaculture research facility made by students enrolled in our Spring 2004 'distance learning course', with John Gonzalez giving the group a first-hand introduction to the challenge of Tilapia growth in relation to STAR residuals post-processing. Over the past year, Dr. Brown's research team has continued their efforts with complementary development of the involved fish species, as well as working on various technical methods for dewatering of the STAR product (e.g., centrifuging, screening, etc.). The underlying concern which has proven to be quite significant is that STAR residuals contained an elevated ammonia level which then mandates separation of the bacterial solids prior to addition to the fish growth chambers. 4) STAR residuals post-processing and water recovery with plants (Jeff Volenec and Brad Joern @ Purdue): After an initial phase of plant growth trials (using various clover and grass options, as well as wheat, tomato, and pepper varieties),the conclusion was reached that all of the test species grown in STAR exhibited reduced growth and enhanced senescence. In turn, a second plant growth study was conduced using another candidate plant group, including: cattail, rush, Kopu II White Clover, Reed Canarygrass, and Super Dwarf Rice. The latter study showed that cattail was the only specie observed to have roots actively growing in STAR effluent, while senescence of roots occurred in STAR effluent for all other species. Upcoming testing will focus on plant growth substrate optimization and species selection 5) STAR residuals post-processing with edible fungi (Caula Beyl): Dr. Beyl's research group has noted that Pleurotus ostreatus 'Grey dover' inoculated on approximately 80g of autoclaved wheat straw supplemented with 0, 20, 40 and 60 % (v/v) dilutions of STAR effluent in 750 ml food containers produced rapid mycelial growth , with approximately an 8cm diameter surface spread within one week. Furthermore, the rate of initial colonization did not appear to be affected by the STAR concentration in the substrate mixture. In fact, experiments using up to 100% concentration (undiluted) of STAR residue in rice straw have been initiated and preliminary data has shown that the undiluted STAR residual is most supportive of mycelial growth in at least three different strains of P. ostreatus, including Pohu, Grey dover and Blue

	dolphin.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Papers from Meeting Proceedings	Glass C, Ressa CW. "Nitrogen Recovery during Solid Waste Treatment in Advanced Life Support." International Conference on Environmental Systems, Colorado Springs, CO, July 2004. International Conference on Environmental Systems, July 2004. Paper 2004-01-2514. , Jul-2004
Papers from Meeting Proceedings	Alleman JE. "Solids Thermophilic Aerobic Reactor for Solid Waste Management in Advanced Life Support Systems." International Conference on Environmental Systems, Colorado Springs, CO, July 2004. International Conference on Environmental Systems, July 2004. Paper 2004-01-2467. , Jul-2004
Presentation	Alleman, J. E. "“Solids Thermophilic Aerobic Reactor for Solid Waste Management in Advanced Life Support Systems”" Habitation Conference, Orlando, FL Jan-2004
Presentation	Glass, Charles and Chee Wah, Ressa, " “Nitrogen Recovery during Solid Waste Treatment in Advanced Life Support Preliminary Results,”" Habitation Jan-2004
Presentation	Alleman, James, Brown, Paul, Glass, Charles, Joern, Brad C., and Volenec, Jeffrey, "“Coordinated Biochemical Stabilization and Resource Recovery with Human-, Plant-, and Food-Derived ALS Solid Wastes,”" Habitation Jan-2004
Presentation	S.M. Howard, J.J. Volenec, B.C. Joern, J.E. Alleman, D.R. Whitaker, "Biosolids Dewatering of Solid-Phase Thermophilic Aerobic Reactor (STAR) Effluent Using Various Plant Species" Soil Science Society of America, Annual Meetings 2004, Jul-2004
Presentation	Alleman, James E., P.B. Brown, C.C. Glass, B.C. Joern and J.C. Volenec. "Coordinated Biochemical Stabilization and Resource Recovery with Human-, Plant-, and Food-Derived ALS Solid Wastes" Habitation 2004, Orlando, FL. Jan-2004
Presentation	Gonzalez, J.M., Y.V. Wu, T. Powless and P.B. Brown. "Evaluation of fish meal-free practical diets for first feeding Nile tilapia Oreochromis niloticus" Aquaculture 2004, Honolulu, HI. Jul-2004
Significant Media Coverage	Alleman JE. "'STAR Project Within ALS' articles appearing in campus and local papers as well as internationally circulated papers such as the New York Times." Television coverage included local news, ABC sports, and CBS news, November 2004., Nov-2004